

Отчёт о лабораторной работе

Лабораторная работа №14

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Цель работы

Получить навыки создания разделов на диске и файловых систем. Получить навыки монтирования файловых систем.

Задание

Научиться создавать разделы на диске и файловые системы, монтировать файловые системы.

Выполнение лабораторной работы

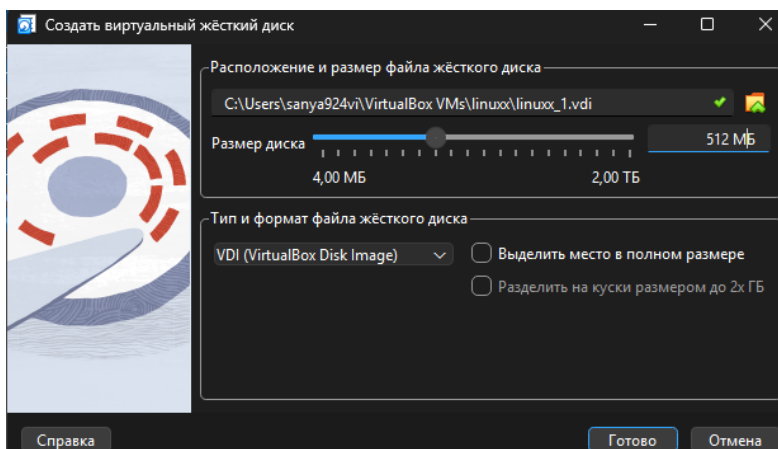


Рис 1. Создаем виртуальный диск

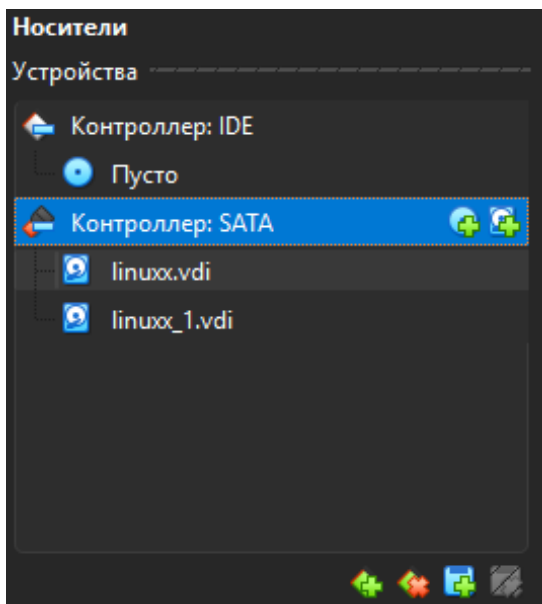


Рис 2. Диск создан

```
adskalozub@localhost:~$ sudo -i
[sudo] пароль для adskalozub:
root@localhost:~# fdisk --list
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sda: 40 GiB, 42949672960 bytes, 83886080 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 1AE3D800-CD7B-437B-B5E6-B804F55A4DB9

Device        Start      End  Sectors  Size Type
/dev/sda1      2048    1230847  1228800  600M EFI System
/dev/sda2    1230848    3327999   2097152    1G Linux extended boot
/dev/sda3    3328000    83884031  80556032  38,4G Linux LVM
```

Рис 3. Проверка дисков

```

root@localhost:~# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xe21aa5d4.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

root@localhost:~# m
bash: m: команда не найдена...
root@localhost:~# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): m

Help:

DOS (MBR)
a  toggle a bootable flag
b  edit nested BSD disklabel
c  toggle the dos compatibility flag

```

Рис 4. Проверка инструментов

```

Command (m for help): p
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4

```

Рис 5. Вывод информации

```

Command (m for help): p
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-1048575, default 2048): 2048
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-1048575, default 1048575): +100M

Created a new partition 1 of type 'Linux' and of size 100 MiB.

```

Рис 6. Создаем раздел размером в 100 Мбайт

```

Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all):
Hex code or alias (type L to list all): 83
Changed type of partition 'Linux' to 'Linux'.

```

Рис 7. изменяем раздел

```

root@localhost:~# fdisk -l /dev/sdb
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4
root@localhost:~# cat /proc/partitions
major minor #blocks name

 8         16    524288 sdb
 8          0 41943040 sda
 8          1   614400 sda1
 8          2   1048576 sda2
 8          3 40278016 sda3
11          0   1048575 sr0
253         0 36081664 dm-0
253         1 4194304  dm-1
root@localhost:~# partprobe /dev/sdb

```

Рис 8. Проверка

```

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): e
Partition number (1-4, default 1): 4
First sector (2048-1048575, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-1048575, default 1048575):

Created a new partition 4 of type 'Extended' and of size 511 MiB.

```

Рис 9. Создаем новый раздел

```

Command (m for help): n
All space for primary partitions is in use.
Adding logical partition 5
First sector (4096-1048575, default 4096):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4096-1048575, default 1048575): +101
M

Created a new partition 5 of type 'Linux' and of size 101 MiB.

```

Рис 10. Создаем новый раздел

```

root@localhost:~# partprobe /dev/sdb
root@localhost:~# cat /proc/partitions
major minor #blocks name

    8         16     524288 sdb
    8         20          0 sdb4
    8         21    103424 sdb5
    8          0  41943040 sda
    8          1    614400 sda1
    8          2   1048576 sda2
    8          3  40278016 sda3
   11          0    1048575 sr0
  253          0  36081664 dm-0
  253          1  4194304 dm-1
root@localhost:~# fdisk --list /dev/sdb
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4

Device      Boot Start      End Sectors  Size Id Type
/dev/sdb4                2048 1048575 1046528  511M  5 Extended
/dev/sdb5                4096  210943  206848  101M 83 Linux

```

Рис 11. проверка

```

root@localhost:~# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
All space for primary partitions is in use.
Adding logical partition 6
First sector (212992-1048575, default 212992):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (212992-1048575, default 1048575): +100M

Created a new partition 6 of type 'Linux' and of size 100 MiB.

```

Рис 12. создание раздела

```

Command (m for help): t
Partition number (4-6, default 6):
Hex code or alias (type L to list all): 82

Changed type of partition 'Linux' to 'Linux swap / Solaris'.

```

Рис 13. Выбираем раздел

```

root@localhost:~# partprobe /dev/sdb
root@localhost:~# cat /proc/partitions
major minor #blocks name

 8      16    524288 sdb
 8      20         0 sdb4
 8      21    103424 sdb5
 8      22    102400 sdb6
 8       0   4194304 sda
 8       1    614400 sda1
 8       2   1048576 sda2
 8       3   40278016 sda3
11       0    1048575 sr0
253      0   36081664 dm-0
253      1    4194304 dm-1
root@localhost:~# fdisk --list /dev/sdb
Disk /dev/sdb: 512 MiB, 536870912 bytes, 1048576 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0xe21aa5d4

Device Boot Start End Sectors Size Id Type
/dev/sdb4      2048 1048575 1046528 511M 5 Extended
/dev/sdb5      4096 210943 206848 101M 83 Linux
/dev/sdb6     212992 417791 204800 100M 82 Linux swap / Solaris
root@localhost:~# mkswap /dev/sdb6
Setting up swapspace version 1, size = 100 MiB (104853504 bytes)
no label, UUID=8e30dac3-1d4a-4b41-b217-7dae55fd6b29
root@localhost:~# swapon /dev/sdb6
root@localhost:~# free -m
              total        used         free      shared  buff/cache   available
Mem:           7660         2182         4336          99        1477        5477
Swap:          4195             0         4195

```

Рис 14. Проверка

```

root@localhost:~# gdisk -l /dev/sdc
bash: gdisk: команда не найдена...

```

Рис 15. Не удалось выполнить этап из-за

```

root@localhost:~# mkfs.xfs /dev/sdb1
Error accessing specified device /dev/sdb1: Нет такого файла или каталога
Usage: mkfs.xfs
/* blocksize */          [-b size=num]
/* config file */        [-c options=xxx]
/* metadata */           [-m crc=0|1,finobt=0|1,uuid=xxx,rmapbt=0|1,reflink=0|1,
                          inobtcount=0|1,bigtime=0|1,autofsck=xxx]
/* data subvol */        [-d agcount=n,agsize=n,file=name=xxx,size=num,
                          (sunit=value,swidth=value|su=num,sw=num|noalign),
                          sectsize=num,concurrency=num]

/* force overwrite */    [-f]
/* inode size */         [-i perblock=n|size=num,maxpct=n,attr=0|1|2,
                          projid32bit=0|1,sparse=0|1,nrxt64=0|1,
                          exchange=0|1]

/* no discard */         [-K]
/* log subvol */         [-l agnum=n,internal,size=num,logdev=xxx,version=n
                          sunit=value|su=num,sectsize=num,lazy-count=0|1,
                          concurrency=num]

/* label */              [-L label (maximum 12 characters)]
/* naming */             [-n size=num,version=2|4,i,ftype=0|1,parent=0|1]]
/* no-op info only */    [-N]
/* prototype file */     [-p fname]
/* quiet */              [-q]
/* realtime subvol */    [-r extsize=num,size=num,rtdev=xxx]
/* sectorsize */         [-s size=num]
/* version */            [-V]
                          devicename

<devicename> is required unless -d name=xxx is given.
<num> is xxx (bytes), xxxs (sectors), xxxb (fs blocks), xxxk (xxx KiB),
      xxxm (xxx MiB), xxxg (xxx GiB), xxxt (xxx TiB) or xxxp (xxx PiB).
<value> is xxx (512 byte blocks).
root@localhost:~# xfs_admin -L xfsdisk /dev/sdb1
/dev/sdb1: Нет такого файла или каталога

fatal error -- couldn't initialize XFS library
root@localhost:~# mkfs.ext4 /dev/sdb5
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 103424 1k blocks and 25896 inodes
Filesystem UUID: 0e694203-b227-4f5f-a1a0-2a1595d727e4
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729

Allocating group tables: done
Writing inode tables: done
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done

```

```

root@localhost:~# tune2fs -L ext4disk /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
root@localhost:~# tune2fs -o acl,user_xattr /dev/sdb5
tune2fs 1.47.1 (20-May-2024)
root@localhost:~# mkdir -p /mnt/tmp
root@localhost:~# mount /dev/sdb5 /mnt/tmp
root@localhost:~# mount
/dev/mapper/rl-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,log
bsize=32k,noquota)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=3894340k,nr_inodes=973585,m
ode=755,inode64)

```

Рис 16. вывод информации

```

root@localhost:~# umount /dev/sdb5
root@localhost:~# mount
/dev/mapper/rl-root on / type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,log
bsize=32k,noquota)
devtmpfs on /dev type devtmpfs (rw,nosuid,seclabel,size=3894340k,nr_inodes=973585,m
ode=755,inode64)

```

Рис 17. вывод информации

```

root@localhost:~# mkdir -p /mnt/data
root@localhost:~# blkid
/dev/mapper/rl-root: UUID="817554fb-cd64-49be-a809-40bc3d466a69" BLOCK_SIZE="512" T
YPE="xfs"
/dev/sda3: UUID="c1n6CR-mvxb-3xtY-48dj-fyfj-v641-iiiBrY" TYPE="LVM2_member" PARTUUI
D="9ecc4106-f8ed-425f-87bb-897416b39054"
/dev/mapper/rl-swap: UUID="b4d4013d-cadd-4b6e-931a-b95022f1bf32" TYPE="swap"
/dev/sdb4: PTTYPE="dos" PARTUUID="e21aa5d4-04"
/dev/sdb5: LABEL="ext4disk" UUID="0e694203-b227-4f5f-a1a0-2a1595d727e4" BLOCK_SIZE=
"1024" TYPE="ext4" PARTUUID="e21aa5d4-05"
/dev/sdb6: UUID="8e30dac3-1d4a-4b41-b217-7dae55fd6b29" TYPE="swap" PARTUUID="e21aa5
d4-06"
/dev/sda2: UUID="830074e3-7303-41c9-9e1a-966908e9dfb7" BLOCK_SIZE="512" TYPE="xfs"
PARTUUID="04832f16-d603-4253-a34b-0a4a7e2dd9be"
/dev/sda1: UUID="EF67-AE5A" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI System Part
ition" PARTUUID="ee6403b4-3a50-4a4d-841e-ed71f60f8435"
root@localhost:~# blkid /dev/sdb1

```

Рис 18. Настройка раздела и проверка

```

root@localhost:~# mount -a
root@localhost:~# df -h

```

Файловая система	Размер	Использовано	Дост	Использовано%	Смонтировано в
/dev/mapper/rl-root	35G	5,1G	30G	15%	/
devtmpfs	3,8G	0	3,8G	0%	/dev
tmpfs	3,8G	84K	3,8G	1%	/dev/shm
efivarfs	256K	51K	201K	20%	/sys/firmware/efi/efiva
rs					
tmpfs	1,5G	1,3M	1,5G	1%	/run
tmpfs	1,0M	0	1,0M	0%	/run/credentials/system
d-journald.service					
/dev/sda2	960M	459M	502M	48%	/boot
/dev/sda1	599M	8,4M	591M	2%	/boot/efi
tmpfs	767M	148K	766M	1%	/run/user/1000
tmpfs	767M	56K	766M	1%	/run/user/0

Рис 19. Настройка и проверка

Рис 20. Создаем директорию и выводим информацию

Рис 21. Выводим информацию по разделам

Вывод

Научились создавать и редактировать разделы дисков

Ответы на контрольные вопросы

1. gdisk или parted
2. используется fdisk
3. файл /etc/fstab
4. в /etc/fstab опция noauto
5. mkswap /dev/имя_раздела
6. mount -a

7. создаётся файловая система ext2
8. `mkfs.ext4 /dev/имя_раздела`
9. `blkid`